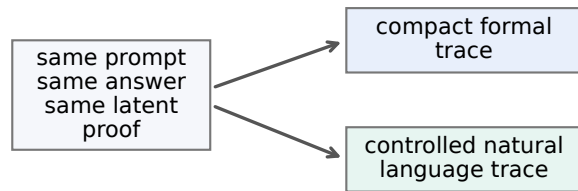


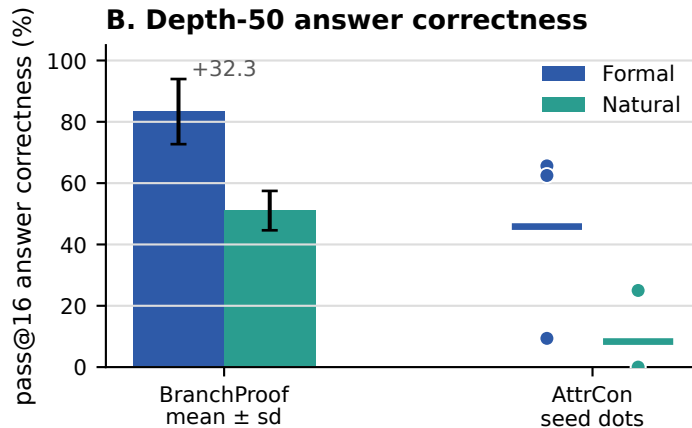
# Formal traces improve extrapolative answers, but proof validity remains a separate target

## A. Controlled substrate comparison

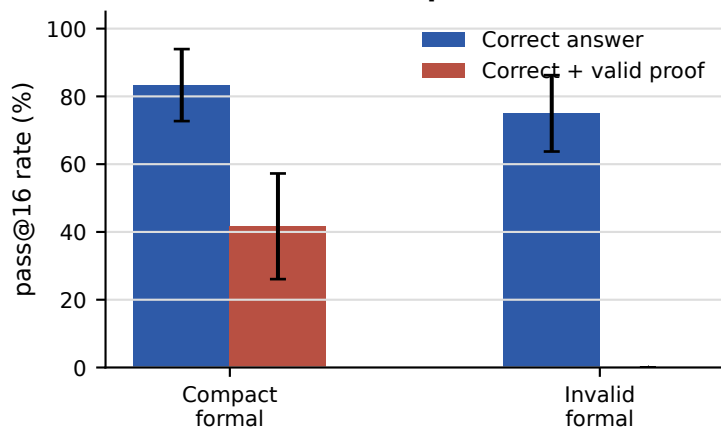


Only the trace substrate changes.

## B. Depth-50 answer correctness



## C. Correctness is not proof faithfulness



## D. Representative sample excerpts

### Formal depth-50 success

```
gold=olive; answer=olive; valid=1
<formal> <constants> a = a b = b c = c
d = d e = e f = f g = g h = h i = i j =
j k = k l = l ...
```

### Natural depth-50 failure

```
gold=juniper; answer=missing; valid=0
<think> <premises> a is orchid. a is
east. If a is south and a is orchid,
then b is cedar. I ...
```

Excerpts come from sample\_generation\_snippets.csv; metrics use 16 samples per prompt.